

Python Assignment 13

1. Write a program which accepts length and width of rectangle and prints area.

Answer:

```
def Rectangle(Value1, Value2):
    Result = Value1 * Value2
    return Result

def main():
    print("Enter the Length : ")
    len = float(input())

    print("Enter the Width : ")
    wid = float(input())

    area = Rectangle(len, wid)

    print("Area of Reactangle is : ", area)

if __name__ == "__main__":
    main()
```

2. Write a program which accepts radius of circle and prints area of circle.

Answer:

```
def Circle(ValueOfRadius):
    pi = 3.14
    R = ValueOfRadius

    #Formula : Area Of Circle = Pi * (Radius*Radius)
    Result = pi * R**2
    return Result

def main():
    Radius =float(input("Enter the value Of Radius : "))
    AreaOfCircle = Circle(Radius)
    print("Area of Circle is : ", AreaOfCircle)

if __name__ == "__main__":
    main()
```

3. Write a program which accepts one number and checks whether it is perfect number or not.

Input: 6

Output: Perfect Number

Answer :

```
def Divisors(Value):
    divs = []
    for i in range(1,Value):
        if Value%i==0:
            divs.append(i)
    return divs

def main():
    iValue = int(input("Enter The Number : "))
    DivisorsList = Divisors(iValue)
    print("Ours Divisors is : ",DivisorsList)

    DivAdd = sum(DivisorsList)
    if DivAdd == iValue:
        print(f"'{iValue} is the Perfect number")
    else:
        print(f"'{iValue} is not a Perfect number")

if __name__ == "__main__":
    main()
```

4. Write a program which accepts one number and prints binary equivalent.

Answer :

```
def DTOB(Value):
    Remainder = []
    while Value > 0:
        remain = Value % 2
        Remainder.append(remain)
        Value = Value // 2
        rem = Remainder[::-1]
    return rem

def main():
```

```

    Decimal = int(input("Enter the Decimal Number : "))
    BinaryList = DTOB(Decimal)
    print(BinaryList)

if __name__ == "__main__":
    main()

```

5. Write a program which accepts marks and displays grade.

Condition Example:

- $\geq 75 \rightarrow$ Distinction
- $\geq 60 \rightarrow$ First Class
- $\geq 50 \rightarrow$ Second Class
- $< 50 \rightarrow$ Fail

Answer:

```

def Grade(Value):
    if Value >= 75 and Value <= 100:
        print("Your Grade is 'Distiction'")

    if Value >= 60 and Value < 75:
        print("Your Grade is '1st Class'")

    if Value >= 50 and Value < 60:
        print("Your Grade is '2nd Class'")

    if Value >= 35 and Value < 50:
        print("Your Grade is '3rd Class'")

    if Value >= 0 and Value < 35:
        print("Your Grade is 'Fail'")

def main():
    Marks = int(input("Enter the marks : "))
    Grade(Marks)

if __name__ == "__main__":
    main()

```